

# Pairwise Consensus Scoring for Robust Kalman Filtering in UWB Indoor Positioning

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**Abstract**—Ultra-wideband (UWB) indoor positioning systems suffer from non-line-of-sight (NLOS) errors that severely degrade localization accuracy. This letter proposes a *Pairwise Consensus* (PC) scoring mechanism that adaptively adjusts the measurement noise covariance of Kalman-family filters by evaluating inter-anchor consistency using a heavy-tailed Student- $t$  kernel. The proposed method is generic and can be integrated into any Kalman filter variant—KF, EKF, or UKF—without requiring explicit NLOS identification, channel models, or machine learning training. Experiments on real UWB ranging data collected in a  $4 \times 8.8$  m indoor environment with four anchors demonstrate that PC scoring consistently improves all three filter variants, reducing RMSE by 13.5%–32.7% under line-of-sight (LOS) conditions. When integrated with UKF, the proposed PC-UKF achieves an RMSE of 89.0 mm in LOS and 129.8 mm under intermittent anchor failure, outperforming the standard UKF, Huber-UKF, and MCC-UKF in both scenarios.

**Index Terms**—Indoor positioning, ultra-wideband, Kalman filter, robust estimation, pairwise consensus, anchor reliability

## I. INTRODUCTION

ULTRA-WIDEBAND (UWB) technology has emerged as a promising solution for centimeter-level indoor positioning due to its fine time resolution and multipath resilience [1]. However, in practical indoor environments, non-line-of-sight (NLOS) conditions—caused by walls, furniture, and human bodies—introduce positive biases in ranging measurements, severely degrading positioning accuracy [2].

Existing NLOS mitigation strategies broadly fall into three categories: *identification-then-mitigation*, where NLOS measurements are detected and either discarded or corrected [3]; *robust estimation*, which employs heavy-tailed cost functions to down-weight outliers [4], [5]; and *machine learning approaches*, which require large training datasets and may not generalize across environments [6].

Robust Kalman filtering variants such as the Huber-UKF [4] and Maximum Correntropy Criterion UKF (MCC-UKF) [5] address outliers by iteratively reweighting innovations based on individual residual magnitudes. However, these methods evaluate each measurement independently and do not exploit the geometric consistency among multiple anchors.

In this letter, we propose a *Pairwise Consensus* (PC) scoring mechanism that assesses the reliability of each anchor by comparing its ranging residual against all other anchors using a Student- $t$  kernel. The resulting consensus score is used to adaptively scale the measurement noise covariance  $R_i$  for each anchor: trusted anchors receive lower  $R_i$  (more influence),

while suspicious anchors are penalized with higher  $R_i$ . This approach is:

- **Filter-agnostic**: applicable to KF, EKF, and UKF with minimal modification;
- **Training-free**: requires no labeled data or environment-specific calibration;
- **Lightweight**: adds only  $\mathcal{O}(N^2)$  pairwise comparisons per timestep, where  $N$  is the number of anchors.

## II. SYSTEM MODEL

### A. UWB Ranging and Positioning

Consider  $N$  anchors at known 2D positions  $\mathbf{a}_i = [a_{x,i}, a_{y,i}]^\top$ ,  $i = 1, \dots, N$ , mounted at a common height  $H$  above the ground plane. The slant-range measurement from anchor  $i$  to a ground-level tag at position  $\mathbf{p} = [x, y]^\top$  is

$$z_i = \sqrt{(x - a_{x,i})^2 + (y - a_{y,i})^2 + H^2} + n_i + b_i, \quad (1)$$

where  $n_i \sim \mathcal{N}(0, \sigma_n^2)$  is measurement noise and  $b_i \geq 0$  is a non-negative NLOS bias.

The ground distance is recovered via  $d_i = \sqrt{z_i^2 - H^2}$ , and the tag position is estimated by least squares (LS) multilateration using anchor 0 as the reference [7]:

$$\hat{\mathbf{p}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}, \quad (2)$$

where  $\mathbf{A}$  and  $\mathbf{b}$  are the linearized geometry matrices.

### B. Kalman Filter Variants

We consider a random-walk dynamic model  $\mathbf{x}_k = \mathbf{x}_{k-1} + \mathbf{w}_k$ ,  $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, q\mathbf{I})$ , where the state is the 2D position  $\mathbf{x}_k = [x_k, y_k]^\top$ .

**KF**: When applied at the range level, independent 1D Kalman filters smooth each anchor's distance series. The filtered distances are then fed to LS for positioning.

**EKF**: Operates directly on the 2D state with the nonlinear observation model  $h_i(\mathbf{x}) = \sqrt{(x - a_{x,i})^2 + (y - a_{y,i})^2 + H^2}$  and its analytical Jacobian  $\mathbf{H}$ .

**UKF**: Uses the unscented transform with sigma points to handle the nonlinearity without requiring Jacobian computation, with parameters  $\alpha = 10^{-3}$ ,  $\beta = 2$ ,  $\kappa = 0$ .

## III. PROPOSED PAIRWISE CONSENSUS SCORING

The core idea is to evaluate each anchor's measurement reliability by comparing its ranging residual with those of other anchors. An anchor whose residual is consistent with the majority is deemed trustworthy; an outlier anchor is penalized.

### A. Consensus Score Computation

At each timestep  $k$ , given raw ground distances  $\mathbf{d}_k = [d_{1,k}, \dots, d_{N,k}]^\top$ :

**Step 1 — Geometric residuals.** Compute a preliminary LS position estimate  $\hat{\mathbf{p}}_k$  from  $\mathbf{d}_k$  using (2). The residual for anchor  $i$  is

$$r_i = |d_{i,k} - \|\hat{\mathbf{p}}_k - \mathbf{a}_i\||. \quad (3)$$

**Step 2 — Pairwise comparison.** For each anchor pair  $(i, j)$ , compute the similarity using a Student- $t$  kernel with  $\nu$  degrees of freedom:

$$c_{ij} = \left(1 + \frac{(r_i - r_j)^2}{\nu\sigma^2}\right)^{-(\nu+1)/2}, \quad (4)$$

where  $\sigma$  is a bandwidth parameter controlling sensitivity. We fix  $\nu = 4$  throughout.

**Step 3 — Consensus score.** The score for anchor  $i$  is the average pairwise agreement:

$$s_i = \frac{1}{N-1} \sum_{j \neq i} c_{ij}, \quad s_i \in [0, 1]. \quad (5)$$

A score  $s_i \approx 1$  indicates that anchor  $i$ 's residual is consistent with all others (trusted), while  $s_i \approx 0$  flags it as an outlier.

### B. Adaptive Measurement Noise

The consensus scores are mapped to per-anchor measurement noise:

$$R_i = R_{\text{base}} \cdot (1 + R_{\text{scale}} \cdot (1 - s_i)). \quad (6)$$

This construction ensures:

- *Trusted anchor* ( $s_i \rightarrow 1$ ):  $R_i \rightarrow R_{\text{base}}$  — the filter fully utilizes the measurement.
- *Outlier anchor* ( $s_i \rightarrow 0$ ):  $R_i \rightarrow R_{\text{base}}(1 + R_{\text{scale}})$  — the measurement is heavily discounted.

### C. Integration with Kalman Filter Variants

**PC-KF:** Each anchor has an independent 1D KF with adaptive  $R_i$  from (6). The predict-update equations are:

$$P_{i,k|k-1} = P_{i,k-1} + Q, \quad (7)$$

$$K_i = \frac{P_{i,k|k-1}}{P_{i,k|k-1} + R_i}, \quad (8)$$

$$\hat{d}_{i,k} = \hat{d}_{i,k-1} + K_i(d_{i,k} - \hat{d}_{i,k-1}). \quad (9)$$

The filtered distances are then passed to LS for 2D positioning.

**PC-EKF:** The adaptive  $R_i$  values form a diagonal matrix  $\mathbf{R}_k = \text{diag}(R_1, \dots, R_N)$  used in the standard EKF update:

$$\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^\top + \mathbf{R}_k, \quad (10)$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^\top \mathbf{S}_k^{-1}, \quad (11)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1})). \quad (12)$$

**PC-UKF:** Identical to PC-EKF but the Jacobian-based linearization is replaced by the unscented transform. The adaptive  $\mathbf{R}_k$  is incorporated into the measurement covariance:

$$\mathbf{P}_{zz,k} = \sum_{i=0}^{2n} W_i^{(c)} (\mathbf{z}_i - \hat{\mathbf{z}}_k)(\mathbf{z}_i - \hat{\mathbf{z}}_k)^\top + \mathbf{R}_k. \quad (13)$$

The overall algorithm is summarized in Algorithm 1.

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### Algorithm 1 Pairwise Consensus Kalman Filter (PC-KF/EKF/UKF)

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**Require:** Raw distances  $\mathbf{d}_k$ , filter state  $\hat{\mathbf{x}}_{k-1}$ ,  $\mathbf{P}_{k-1}$

**Ensure:** Updated state  $\hat{\mathbf{x}}_k$ ,  $\mathbf{P}_k$ , scores  $\mathbf{s}_k$

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1:  $\hat{\mathbf{p}}_k \leftarrow \text{LS}(\mathbf{d}_k)$  {Preliminary position}
2: for  $i = 1$  to  $N$  do
3:    $r_i \leftarrow |d_{i,k} - \|\hat{\mathbf{p}}_k - \mathbf{a}_i\||$  {Residual}
4: end for
5: for  $i = 1$  to  $N$  do
6:    $s_i \leftarrow \frac{1}{N-1} \sum_{j \neq i} \left(1 + \frac{(r_i - r_j)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}$ 
7: end for
8: for  $i = 1$  to  $N$  do
9:    $R_i \leftarrow R_{\text{base}} \cdot (1 + R_{\text{scale}} \cdot (1 - s_i))$ 
10: end for
11:  $\mathbf{R}_k \leftarrow \text{diag}(R_1, \dots, R_N)$ 
12: Perform KF/EKF/UKF predict step
13: Perform KF/EKF/UKF update with  $\mathbf{R}_k$ 
14: return  $\hat{\mathbf{x}}_k$ ,  $\mathbf{P}_k$ ,  $\mathbf{s}_k$ 

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## IV. EXPERIMENTAL RESULTS

### A. Setup

The experimental testbed consists of four DW1000-based UWB anchors mounted at height  $H = 1400$  mm at the corners of a  $4000 \times 8800$  mm rectangular room, with coordinates  $\mathbf{a}_1 = [4000, 8800]$ ,  $\mathbf{a}_2 = [0, 8800]$ ,  $\mathbf{a}_3 = [0, 0]$ , and  $\mathbf{a}_4 = [4000, 0]$  (mm). The tag is moved along a rectangular path with waypoints  $A(800, 400) \rightarrow B(3000, 400) \rightarrow C(3000, 8000) \rightarrow D(800, 8000) \rightarrow A$  at an approximate speed of 200 mm/s, yielding a total path length of 19,600 mm.

Data is collected in two scenarios: (i) **LOS**: eight independent runs where all four anchors maintain line-of-sight to the tag, with ranging at approximately 60 Hz; and (ii) **Degraded anchor**: one run where a single anchor experiences intermittent ranging failures—alternating between valid measurements and dropouts.

### B. PC Enhancement Across Filter Variants

Table I presents the LOS positioning accuracy of each baseline filter and its PC-enhanced counterpart. The PC mechanism consistently improves all metrics across all three filter variants:

- **KF**  $\rightarrow$  **PC-KF**: RMSE reduced from 285.6 to 192.3 mm (−32.7%);
- **EKF**  $\rightarrow$  **PC-EKF**: RMSE reduced from 112.4 to 97.2 mm (−13.5%);
- **UKF**  $\rightarrow$  **PC-UKF**: RMSE reduced from 106.5 to 89.0 mm (−16.4%).

The improvement is most pronounced for the KF variant, which operates at the range level and directly benefits from per-anchor noise adaptation. The 2D filters (EKF, UKF) already capture geometric constraints, yet PC scoring still provides meaningful gains by further down-weighting unreliable anchor measurements.

TABLE I: LOS Positioning Accuracy: Baseline vs. PC-Enhanced Filters (mm)

| Method      | RMSE         | MAE          | CEP50        | P95          | MAX          |
|-------------|--------------|--------------|--------------|--------------|--------------|
| Raw + LS    | 342.1        | 241.7        | 169.6        | 748.4        | 1713.2       |
| KF + LS     | 285.6        | 212.8        | 160.0        | 668.9        | 852.1        |
| PC-KF + LS  | <b>192.3</b> | <b>148.6</b> | <b>117.7</b> | <b>389.8</b> | <b>542.6</b> |
| Improvement | 32.7%        | 30.2%        | 26.4%        | 41.7%        | 36.3%        |
| EKF-2D      | 112.4        | 88.2         | 71.3         | 229.2        | 317.2        |
| PC-EKF-2D   | <b>97.2</b>  | <b>76.2</b>  | <b>64.9</b>  | <b>202.6</b> | <b>304.1</b> |
| Improvement | 13.5%        | 13.6%        | 9.0%         | 11.6%        | 4.1%         |
| UKF-2D      | 106.5        | 83.7         | 66.5         | 218.6        | 309.6        |
| PC-UKF-2D   | <b>89.0</b>  | <b>68.7</b>  | <b>55.5</b>  | <b>193.6</b> | <b>260.1</b> |
| Improvement | 16.4%        | 17.9%        | 16.5%        | 11.4%        | 16.0%        |

TABLE II: LOS Comparison with others Robust UKF Methods (mm)

| Method                   | RMSE        | MAE         | CEP50       | P95          | MAX          |
|--------------------------|-------------|-------------|-------------|--------------|--------------|
| Raw + LS                 | 342.1       | 241.7       | 169.6       | 748.4        | 1713.2       |
| Standard UKF             | 106.5       | 83.7        | 66.5        | 218.6        | 309.6        |
| Huber-UKF [4]            | 106.9       | 83.5        | 68.4        | 221.4        | 362.3        |
| MCC-UKF [5]              | 96.8        | 77.0        | 63.0        | 208.0        | 308.9        |
| <b>PC-UKF (proposed)</b> | <b>89.0</b> | <b>68.7</b> | <b>55.5</b> | <b>193.6</b> | <b>260.1</b> |

### C. Comparison with others Robust Filters

Table II compares the proposed PC-UKF against state-of-the-art robust UKF variants under LOS conditions. All methods use the same UKF core with individually tuned hyperparameters.

The proposed PC-UKF achieves the best performance across all metrics with an RMSE of 89.0 mm, reducing error by 16.4% over the standard UKF, 16.7% over Huber-UKF, and 8.1% over MCC-UKF. The key advantage of PC-UKF is its *pairwise* evaluation strategy: rather than assessing each measurement in isolation (as Huber and MCC do), it exploits geometric redundancy by cross-checking inter-anchor consistency.

### D. Case Study: Intermittent Anchor Failure

To evaluate robustness under adverse conditions, we consider a scenario where one anchor experiences intermittent ranging failures—alternating between valid measurements and dropouts—while the remaining three anchors operate normally. Fig. 1 illustrates the anchor distance measurements and the resulting trajectories.

Table III summarizes the positioning accuracy under this condition. The degraded anchor’s intermittent dropouts cause its ranging residual to deviate sharply from the other anchors, which the pairwise consensus mechanism naturally detects and penalizes via a low consensus score.

PC-UKF achieves 129.8 mm RMSE—a 22.4% improvement over the standard UKF (167.3 mm) and an 11.9% improvement over MCC-UKF (147.4 mm). Compared to the LOS scenario (Table II), the relative advantage of PC-UKF over the standard UKF increases from 16.4% to 22.4%, demonstrating that the pairwise consensus mechanism is particularly effective when

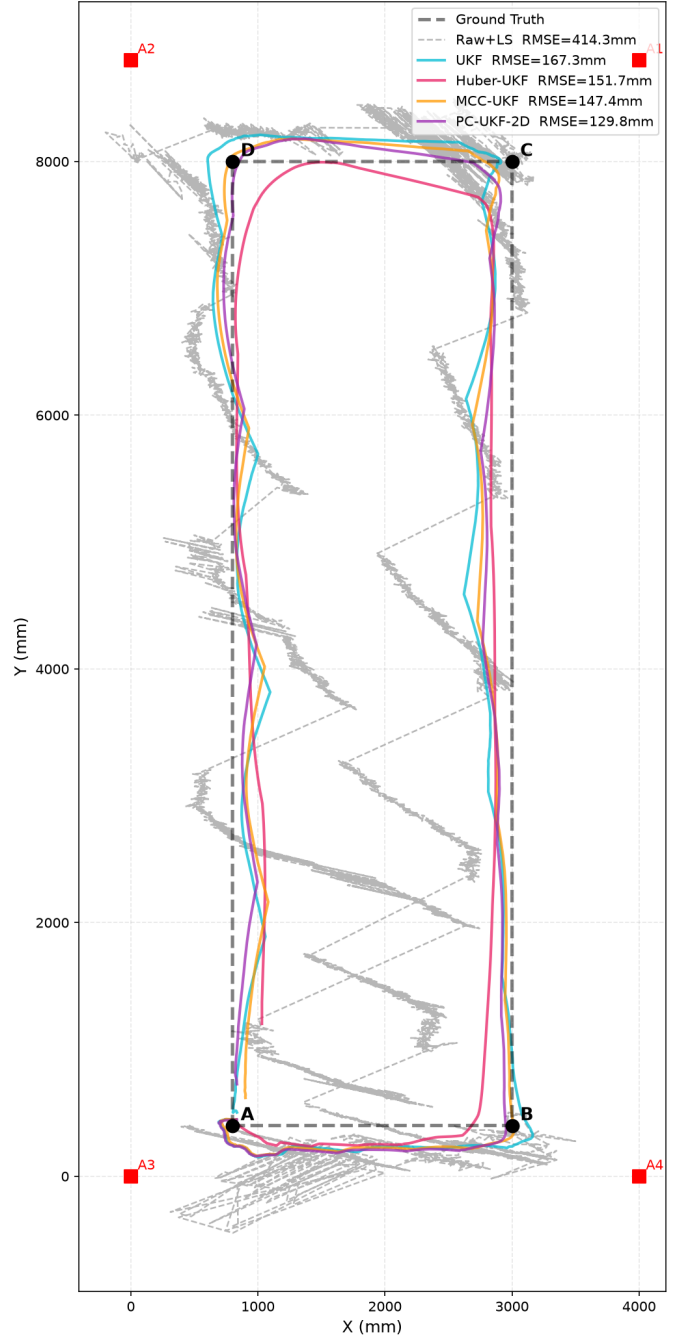


Fig. 1: Intermittent anchor failure case study.

one anchor provides degraded measurements. All robust methods substantially outperform the standard UKF, confirming the need for adaptive measurement noise handling in practical deployments.

### E. Computational Overhead

The PC scoring adds an LS computation ( $\mathcal{O}(N)$ ) and  $N(N-1)/2$  pairwise kernel evaluations per timestep. Measured on a standard CPU, the per-sample processing times are: UKF 0.093 ms, MCC-UKF 0.126 ms, Huber-UKF 0.149 ms, and PC-UKF 0.162 ms. The PC-UKF is  $1.7\times$  slower than the standard UKF but comparable to the iterative methods, making

TABLE III: Degraded Anchor Scenario: SOTA Comparison (mm)

| Method                   | RMSE         | MAE          | CEP50        | P95          | MAX          |
|--------------------------|--------------|--------------|--------------|--------------|--------------|
| Raw + LS                 | 414.3        | 326.0        | 252.7        | 890.4        | 1099.1       |
| Standard UKF             | 167.3        | 145.4        | 154.0        | 295.4        | 378.2        |
| Huber-UKF [4]            | 151.7        | 136.8        | 144.9        | 246.6        | 277.1        |
| MCC-UKF [5]              | 147.4        | 129.2        | 121.1        | 252.7        | 312.3        |
| <b>PC-UKF (proposed)</b> | <b>129.8</b> | <b>112.6</b> | <b>105.1</b> | <b>216.5</b> | <b>247.0</b> |

it practical for real-time applications at typical UWB update rates (10–200 Hz).

## V. CONCLUSION

This letter presented a Pairwise Consensus scoring mechanism for adaptive measurement noise covariance in Kalman-family filters applied to UWB indoor positioning. By leveraging inter-anchor residual consistency through a Student- $t$  kernel, the proposed method reliably identifies and down-weights unreliable measurements without requiring channel models or training data. Experimental results on real UWB data demonstrated consistent improvements across KF, EKF, and UKF variants. Under LOS conditions, PC-UKF achieves 89.0 mm RMSE, outperforming Huber-UKF by 16.7% and MCC-UKF by 8.1%. Under intermittent anchor failure, the advantage is even more pronounced: PC-UKF reduces RMSE by 22.4% over standard UKF and 11.9% over MCC-UKF (129.8 vs. 167.3 and 147.4 mm, respectively). Future work will explore extensions to dynamic environments with more anchors and integration with inertial measurement units.

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